

# **GATI SHAKTI VISHWAVIDYALAYA**

## **C Programming Language**

### **Lab Assignment – 4**

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#### *Instructions:*

1. *Write all the program and execute.*
  2. *Maintain soft copy of all codes and prepare a Lab Record consisting hand written programs (10 Programs out of this list).*
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1. Write a program to find average of all the elements of a 2 dimensional array.
2. Write a program to
  - a) Add 2D array and print
  - b) Subtract one 2D array from another and print it
  - c) Multiply two 2D arrays and print the resultant array.
  - d) Perform transpose of a 2D array.
3. Write a program that reads a matrix and displays
  - a) Sum of its diagonals.
  - b) Sum of the elements above the main diagonal.
4. Write a program to read a string using scanf(), gets(), getchar(), getch() and getche() and correspondingly print it using the respective functions.
5. Write a program to find the position of NULL char in the string array.
6. Write a program to
  - a) compare two strings
  - b) concatenate two strings
  - c) convert all the lowercase char of the string into uppercase.
7. Write a program to check whether a string is palindrome or not.
8. Write a program to extract last n characters of a string.
9. Write a program to extract a substring from a given string where the starting position and length of substring will be given by the user.
10. Write a program to insert a substring into a main string. Main string, substring and position of insertion should be taken as input from user.
11. Write a program to delete a substring from a given string.
12. Write a program to convert a given string of numbers into integer both by using atoi() and also writing its code by own.
13. Write a program to use character manipulation functions i.e isalnum(), isdigit(), islower(), tolower(), isprint(). (Defined in <ctype.h>)

14. Write a program to illustrate the use of following string manipulation functions : strcat(), strcmp(), strcpy(), strncat(), strncmp(), strncpy(), strlen(), strstr() {Except also study the other inbuilt string functions also.}
15. Write a program to read a sentence and count the number of characters in it. (Space should not be counted).